ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ
BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
(Autonomous Institution Under VTU)
Yelahanka, Bengaluru -560119

Department of MCA

(Accredited by NBA, New Delhi)

PROJECT WORK - PHASE I - MMCA34(2025–27 Batch)
Synopsis Evaluation

Student Name	D K Karthikesh	**USN**	1BY25MC034
Project Title	Privacy-Preserving Federated Learning for Social Media Influencer Engagement Prediction and Personalized Content Recommendation		
Project Guide	Prof Reshma C R		

Criteria	Excellent (8–10)	Good (5–7)	Needs Improvement (1–4)
Overall Synopsis Quality (Problem, Literature Survey, Objectives, Methodology, Timeline)	Well-defined problem, good literature review, clear objectives, sound methodology, realistic timeline	Covers all points but lacks depth or clarity in some areas	Problem/objectives vague; methodology or timeline unclear or missing

Component	Maximum Marks	Marks Obtained
Synopsis Evaluation	**10**	

Project Guide

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1.INTRODUCTION

1.1 Background and Context

Social media platforms generate large volumes of interaction data through likes, comments, shares, saves, watch time, clicks, follows, and other engagement signals. For influencers and content creators, predicting how audiences will respond to a post can improve content planning, timing, and recommendation quality. However, conventional machine learning approaches commonly centralize user interaction data, creating privacy risks because behavioural information may contain sensitive preferences, interests, and usage patterns.

Federated Learning (FL) provides an alternative by allowing multiple participating clients to train a shared model locally while transmitting model updates rather than raw engagement records. This project combines federated learning with privacy-preserving mechanisms and an engagement prediction model to reduce exposure of individual-level social media data. The resulting global knowledge can then support personalized content recommendation without requiring a central repository of raw user interactions.

1.2 Project Overview

The proposed system will be a privacy-aware machine learning framework for predicting social media influencer engagement and generating personalized content recommendations. Each client represents a creator, user group, or simulated social-media environment containing local engagement information. A local model learns patterns from private data, and only model parameters or protected updates are communicated to the federated server. The server performs secure aggregation and produces an improved global model that is redistributed to participating clients.

The recommendation component will use predicted engagement, content characteristics, and local preference patterns to rank candidate posts. The framework will compare centralized training with federated training and will evaluate privacy-aware performance using prediction accuracy, precision, recall, F1-score, MAE/RMSE where applicable, recommendation ranking metrics, communication rounds, and training convergence.

1.3 Significance of the Proposed Work

The project addresses a practical tension between personalization and privacy. Instead of treating privacy as a separate afterthought, privacy protection is incorporated into the learning workflow. This makes the system relevant to creator analytics, personalized feeds, advertising intelligence, and other recommendation settings in which behavioural data is valuable but direct centralization is undesirable.

2.LITERATURE SURVEY

2.1 Review of State-of-the-Art Solutions

A review of federated learning, privacy-preserving machine learning, engagement prediction, and recommender-system approaches will be conducted to identify technical strengths and limitations:

Table No: 3. Literature Survey Summary

Ref. No.	Titles	Authors	Contributions	Gap identified
[1]	Communication-Efficient Learning of Deep Networks from Decentralized Data	McMahan et al. (2017)	Introduced federated learning for decentralized model training without centralizing raw data.	Does not specifically address social-media influencer engagement or personalized recommendation.
[2]	Advances and Open Problems in Federated Learning	Kairouz et al. (2021)	Reviews FL challenges including privacy, communication, optimization, and security.	General FL focus; social-media recommendation and influencer analytics are not the main application.
[3]	Practical Secure Aggregation for Privacy-Preserving Machine Learning	Bonawitz et al. (2017)	Proposed secure aggregation so the server can obtain aggregate updates without seeing individual client updates.	Focuses on secure aggregation rather than engagement prediction and recommendation quality.
[4]	The Algorithmic Foundations of Differential Privacy	Dwork & Roth (2014)	Established foundations and mechanisms for protecting individual privacy through controlled noise.	Does not provide a federated social-media recommendation framework or application-specific evaluation.
[5]	Deep Learning Based Recommender System: A Survey and New Perspectives	Zhang et al. (2019)	Reviews deep-learning approaches for personalized recommendation and ranking.	Primarily assumes access to interaction data; privacy-preserving FL is not integrated into the recommendation workflow.
[6]	Deep Learning with Differential Privacy	Abadi et al. (2016)	Proposed DP-SGD, adding calibrated noise and gradient clipping to enable differentially private model training.	General DP training method; not tailored to federated social-media engagement prediction.
[7]	Federated Learning: Challenges, Methods, and Future Directions	Li et al. (2020)	Surveys statistical and systems challenges in FL, including client heterogeneity and communication efficiency.	Survey-level treatment; does not implement a domain-specific engagement or recommendation system.
[8]	Federated Optimization in Heterogeneous Networks (FedProx)	Li et al. (2020)	Proposed FedProx to handle non-IID client data and systems heterogeneity during aggregation.	Optimization-focused; not evaluated on social-media engagement or recommendation tasks.
[9]	Neural Collaborative Filtering	He et al. (2017)	Proposed a neural-network-based collaborative filtering approach for recommendation.	Assumes centralized access to interaction data; no privacy-preserving or federated treatment.
[10]	Deep Neural Networks for YouTube Recommendations	Covington et al. (2016)	Describes a large-scale deep learning recommendation pipeline for candidate generation and ranking.	Centralized industrial system; does not address data privacy or federated training.
[11]	Federated Machine Learning: Concept and Applications	Yang et al. (2019)	Formalizes federated learning categories (horizontal, vertical, transfer) and application scenarios.	Conceptual framework; lacks a concrete engagement-prediction or recommendation implementation.

Ref. No.	Titles	Authors	Contributions	Gap identified
[12]	A Hybrid Approach to Privacy-Preserving Federated Learning	Truex et al. (2019)	Combines differential privacy with secure multiparty computation for stronger FL privacy guarantees.	Focuses on privacy-mechanism design rather than social-media recommendation performance.
[13]	XGBoost: A Scalable Tree Boosting System	Chen & Guestrin (2016)	Introduced an efficient, scalable gradient boosting algorithm widely used for tabular prediction tasks.	General-purpose learner; not evaluated in a federated or privacy-preserving setting.
[14]	LightGBM: A Highly Efficient Gradient Boosting Decision Tree	Ke et al. (2017)	Proposed a faster gradient boosting implementation using histogram-based learning.	General-purpose learner; does not address federated training or engagement prediction directly.
[15]	Flower Federated Learning Documentation	Flower Team (2024)	Provides an open-source framework for orchestrating federated clients, rounds, and aggregation strategies.	A software toolkit rather than a research contribution to engagement prediction or recommendation.
[16]	PyTorch Documentation	PyTorch Team (2024)	Provides the deep-learning library used to implement and train the prediction models.	Implementation tool only; no domain-specific engagement or recommendation content.
[17]	Opacus: Training PyTorch Models with Differential Privacy	Opacus Team (2024)	Provides a library for applying DP-SGD and gradient clipping to PyTorch models.	Generic DP tool; requires integration with the federated engagement-prediction pipeline.
[18]	Scikit-learn User Guide	Scikit-learn Developers (2024)	Provides baseline machine-learning models and preprocessing utilities used for the centralized comparison.	Centralized-learning library; no federated or privacy-preserving capability.
[19]	Sustainable Development Goals: Goal 9, Goal 16 and Goal 8	United Nations (2023)	Defines the global development goals used to justify the project's SDG alignment.	Policy framework, not a technical machine-learning contribution.

- **1. Centralized Social Media Prediction:** Conventional engagement prediction collects user and content interaction data at a central server. Such systems can achieve strong predictive performance but increase the consequences of data breaches and raise concerns regarding user consent and data ownership.
- **2. Federated Learning:** Federated learning distributes model training to participating clients and aggregates local model updates. It reduces the need to transfer raw data, although model updates may still leak information if they are not adequately protected.
- **3. Secure Aggregation:** Secure aggregation protocols allow a server to obtain an aggregate update without directly observing each client's individual contribution. This strengthens confidentiality while retaining the collaborative training process.
- **4. Differential Privacy:** Differential privacy can add calibrated noise to model updates or outputs to reduce the possibility of inferring information about individual records. The privacy-utility trade-off must be evaluated carefully.

- **5. Recommendation Systems:** Content-based, collaborative, and hybrid recommender systems can rank content using historical interactions and content features. Combining these methods with federated learning can support personalization while keeping local preference data decentralized.

2.2 Comparative Summary: Existing Systems vs. Proposed Framework

Table No: 4. Comparative Analysis of Existing Approaches vs. Proposed Framework

Feature / Parameter	Centralized ML	Basic Federated Learning	Proposed Framework
Raw Data Handling	Central server storage	Remains at clients	Remains at clients; privacy-first design
Model Training	Centralized	Distributed local training	Federated local training + secure aggregation
Update Protection	Application dependent	May be exposed	Protected aggregation / optional DP
Engagement Prediction	Supported	Supported	Supported with privacy-aware evaluation
Personalization	High with centralized data	Client-level possible	Local preference-aware recommendations
Privacy Risk	Higher central exposure	Reduced raw-data exposure	Reduced exposure + update protection
Evaluation	Accuracy focused	Accuracy + convergence	Prediction + ranking + privacy/communication trade-off

3.PROBLEM STATEMENT WITH AIM AND OBJECTIVES

3.1 Problem Statement

Social media engagement prediction and content recommendation depend heavily on user interaction data. Existing centralized approaches require large collections of behavioural records, which may expose sensitive user preferences and create a single high-value target for data misuse or breaches. At the same time, creators and platforms need accurate predictions of engagement and relevant personalized recommendations. A practical framework is therefore required that can collaboratively learn from distributed social-media interaction data while reducing the need to share raw records and explicitly evaluating the effect of privacy protection on predictive and recommendation performance.

3.2 Project Aim

To design, develop, and evaluate a privacy-preserving federated learning framework for social media influencer engagement prediction and personalized content recommendation, where distributed clients collaboratively train machine learning models without directly sharing raw user interaction data.

3.3 Project Objectives

- **1. Distributed Data Preparation:** Create or use partitioned social-media engagement datasets representing independent clients and preprocess numerical, categorical, temporal, and content-related features.
- **2. Engagement Prediction Model:** Develop a machine learning model to estimate engagement outcomes such as likes, comments, shares, clicks, or an engagement score from content and historical interaction features.
- **3. Federated Training:** Implement client-side local training and server-side aggregation across multiple communication rounds, comparing federated learning with a centralized baseline.
- **4. Privacy Protection:** Integrate secure aggregation concepts and, where feasible, differential privacy/noise mechanisms to reduce the risk of exposing individual client contributions.
- **5. Personalized Recommendation:** Build a recommendation module that ranks candidate content for a user or audience segment using predicted engagement and local preference information.
- **6. Performance and Privacy Evaluation:** Measure predictive performance, recommendation quality, convergence, communication overhead, and privacy-utility trade-offs under different experimental settings.
- **7. Demonstration Dashboard:** Provide a simple application/dashboard showing client training status, engagement predictions, recommended content, evaluation metrics, and privacy-aware federated learning progress.

4.SDG ALIGNMENT

4.1 UN Sustainable Development Goals (SDG) Alignment

The proposed framework contributes to responsible and trustworthy digital technologies by addressing privacy, secure data use, and equitable access to machine-learning capabilities. Its strongest alignment is with SDG 9 and SDG 16, with supporting relevance to SDG 8:

- **Primary Alignment: SDG 9 — Industry, Innovation and Infrastructure:** The project will develop an innovative distributed AI architecture that enables collaborative machine learning across decentralized data sources and supports resilient digital infrastructure.
- **Secondary Alignment: SDG 16 — Peace, Justice and Strong Institutions:** Privacy-preserving learning supports responsible handling of digital information and encourages transparent, accountable use of data-driven systems.
- **Supporting Alignment: SDG 8 — Decent Work and Economic Growth:** Better analytics and personalized content tools will support creators, digital entrepreneurs, and online businesses through data-driven decision support without requiring unnecessary centralization of personal interaction records.

4.2 ESG Compliance Summary

Table No: 5. Environmental, Social, and Governance (ESG) Framework Mapping

ESG Pillar	Project Contribution	Supporting Evidence / Verification
Environmental (E)	Primarily software based; reduces the need for repeated centralized data movement and can use shared computational infrastructure efficiently.	Communication-round logs; training resource comparison; deployment configuration.
Social (S)	Protects user interaction privacy while enabling personalized content and creator analytics.	Privacy-risk analysis; client data-locality checks; recommendation quality evaluation.
Governance (G)	Uses explicit privacy controls, federated participation, auditable aggregation rounds, and configurable privacy parameters.	Training logs; aggregation records; parameter configuration; experiment reports.

5.METHODOLOGY

5.1 Privacy-Preserving Federated Architecture & System Workflow

The methodology is organized as a distributed learning pipeline in which raw engagement records remain with their respective clients. The system contains four tightly coupled layers:

- **1. Client Data & Feature Layer:** Each client stores local engagement records and derives features such as content type, posting time, historical engagement, audience interaction patterns, and content metadata.
- **2. Local Model Training Layer:** A local engagement prediction model will be trained for a limited number of epochs using only the client's private dataset. Model performance will be measured locally before the update is transmitted.
- **3. Privacy & Aggregation Layer:** Model updates are protected using secure aggregation principles and, where implemented, differential-privacy noise or clipping. The federated server combines protected updates to form a new global model.
- **4. Recommendation & Evaluation Layer:** The global model will be evaluated and distributed back to clients. Predicted engagement is combined with local preference information to rank candidate content and generate personalized recommendations.

5.2 System Data Flow Pipeline

The end-to-end pipeline functions as a repeated feedback loop: Local Dataset → Feature Preprocessing → Local Model Training → Model Update Protection → Secure Aggregation → Global Model → Client Update → Engagement Prediction → Personalized Ranking → Evaluation → Next Federated Round.

5.3 Experimental Procedure & Evaluation

Experiments will compare a centralized baseline with federated configurations across multiple clients and communication rounds. Prediction will be evaluated using Accuracy, Precision, Recall, F1-score and/or MAE/RMSE as appropriate; recommendation quality using Precision@K, Recall@K, NDCG@K and Hit Rate; and federated performance using loss, convergence rounds, training time and communication cost. Privacy will be assessed through raw-data locality, update protection, and differential-privacy parameters such as noise level or privacy budget where implemented.

5.4 System Architecture and Workflow

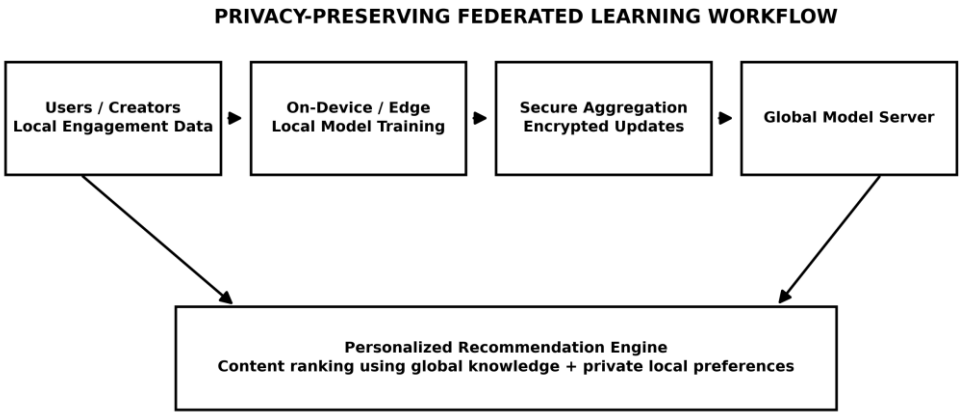


Figure 1: Privacy-Preserving Federated Learning System Pipeline

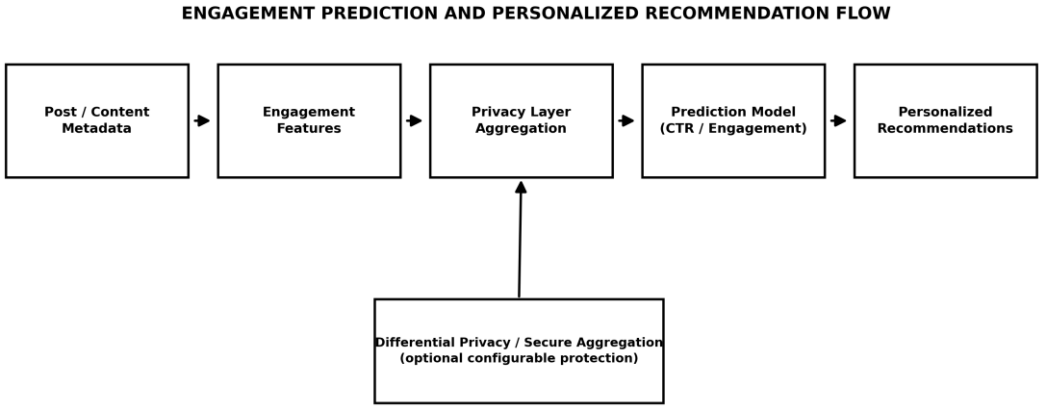


Figure 2: Engagement Prediction and Personalized Recommendation Flow

5.5 Algorithms Specified

The following specific algorithms are identified for implementation at each stage of the pipeline:

- **1. Local engagement prediction model:** A gradient-boosted tree model (XGBoost / LightGBM Regressor-Classifier) will be used as the primary local learner, with a Multi-Layer Perceptron (MLP) neural network evaluated as an alternative for capturing non-linear feature interactions.
- **2. Federated optimization/aggregation algorithm:** Federated Averaging (FedAvg) (McMahan et al., 2017) will be used to aggregate client model updates into the global model; FedProx will be evaluated as an alternative to handle client heterogeneity (non-IID data).
- **3. Secure aggregation:** A secure aggregation protocol based on Bonawitz et al. (2017) will be used so that the server can only access the aggregated sum of client updates and not any individual client's contribution.

- **4. Differential privacy mechanism:** DP-SGD (Differentially Private Stochastic Gradient Descent, Abadi et al., 2016) with per-update gradient clipping and calibrated Gaussian noise addition will be used to bound the privacy budget (ε, δ) during local training.
- **5. Personalized recommendation/ranking algorithm:** Candidate content will be ranked using a pointwise learning-to-rank approach that combines the predicted engagement score with content-similarity scores (cosine similarity over content embeddings), with Neural Collaborative Filtering (He et al., 2017) considered as an extension.

6.MODULES IDENTIFIED

6.1 Functional System Modules

The proposed architecture will be partitioned into seven cohesive and loosely coupled functional modules:

- **Module 1: Data Preparation & Client Partitioning:** Will clean the dataset, encode categorical features, normalize numerical attributes, create train/test splits, and partition records into independent clients.
- **Module 2: Local Engagement Prediction:** Will train a local machine learning model on each client's private data and produce engagement predictions without uploading raw interaction records.
- **Module 3: Federated Learning Coordinator:** Will manage client selection, training rounds, local epochs, model distribution, and global aggregation using a federated learning framework.
- **Module 4: Privacy Protection:** Will apply update clipping, secure aggregation concepts, and optional differential-privacy noise to reduce leakage from individual client contributions.
- **Module 5: Personalized Recommendation Engine:** Will generate candidate content scores from predicted engagement and combine them with client-local preference signals to produce ranked recommendations.
- **Module 6: Evaluation & Experiment Manager:** Will compare centralized and federated configurations using prediction, recommendation, convergence, communication, and privacy-utility metrics.
- **Module 7: Dashboard & Reporting:** Will display model performance, federated rounds, client participation, engagement predictions, recommended content, and experiment summaries in an understandable interface.

7.TOOLS AND TECHNOLOGIES IDENTIFIED

7.1 Software & Framework Stack

- **Python 3.10+:** Primary programming language for data preprocessing, model development, federated training, evaluation, and backend services.
- **PyTorch / TensorFlow:** Deep-learning framework for implementing the engagement prediction model and training routines.
- **Flower / Federated Learning Framework:** Used to orchestrate simulated or real federated clients, communication rounds, and aggregation experiments.
- **Scikit-learn & Pandas:** Used for preprocessing, baseline machine-learning models, feature engineering, data manipulation, and evaluation utilities.
- **NumPy:** Numerical computation and efficient feature/model operations.
- **Opacus or Equivalent Differential Privacy Library:** Optional component for clipping and differential-privacy noise during model training where supported.
- **Flask / FastAPI:** Lightweight backend API layer for exposing predictions, recommendation results, and experiment information.
- **React.js / HTML5 / CSS3:** Front-end technologies for a simple dashboard and visualization of results.
- **Matplotlib / Plotly:** Visualization of convergence curves, metric comparisons, client participation, and privacy-utility trade-offs.
- **Git & GitHub:** Version control, experiment tracking, documentation, and collaborative development.

7.2 Hardware / Execution Environment

The prototype will initially be implemented on a standard laptop or desktop for simulation of multiple federated clients. Optional edge devices can be used later to demonstrate geographically or physically distributed clients. The initial Phase-I implementation will prioritize software simulation so that experiments remain reproducible and low cost.

7.3 Dataset and Data Representation

The project will use a suitable social-media engagement dataset containing content attributes and interaction outcomes. If a real dataset is unavailable, a public benchmark or synthetically partitioned dataset can be used to simulate multiple privacy-separated clients. Raw client partitions will remain isolated during federated experiments.

8.TIMELINE USING GANTT CHART

8.1 Project Schedule and Milestones (Phase-I and Phase-II)

The project lifecycle spans two academic semesters and is organized into six structured phases from problem definition and literature study to federated implementation, evaluation, and final reporting:

Table No: 6. Project Implementation Timeline and Milestone Schedule

Phase / Milestone Activity	M1	M2	M3	M4	M5	M6
Phase 1: Literature Survey & Problem Definition	**XXXX**					
Phase 2: SRS, System Design & Architecture		**XXXX**	**XXXX**			
Phase 3: Dataset Preparation & Baseline Model			**XXXX**	**XXXX**		
Phase 4: Federated Learning & Privacy Integration				**XXXX**	**XXXX**	
Phase 5: Recommendation Engine & Evaluation					**XXXX**	**XXXX**
Phase 6: Testing, Documentation & Final Report						**XXXX**

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